

exp_id	group	description	loss	optimizer	lr	weight_decay	batch	epochs	hidden	dropout	clip_norm	precision	init	seed	step0_loss	best_val_loss	best_epoch	final_train_loss	final_val_loss	val_acc	val_macro_f1	time_per_epoch_s	peak_memory_mb	diverged	eval_acc	eval_macro_f1	figure_file	notes	step0_gap_vs_lnc	gap_val_minus_train	delta_val_f1_vs_base	beyond_noise
base-s1	baseline	Baseline M-base	ce	sgd_momentum	0,1	0	512	20	256-128	0 none	fp32	he		1	2,2691	0,2382	19	0,2168	0,2408	0,9035	0,8520	3,190986		FALSE	0,9016	0,8514	figures/base-s1.png	GPU-memory=N/A on CPU.	0,3232	0,0240	-0,0029	Không
base-s2	baseline	Baseline M-base	ce	sgd_momentum	0,1	0	512	20	256-128	0 none	fp32	he		2	1,9782	0,2245	20	0,2047	0,2245	0,9101	0,8569	3,4998335		FALSE			figures/base-s2.png	GPU-memory=N/A on CPU.	0,0322	0,0198	0,0020	Không
base-s3	baseline	Baseline M-base	ce	sgd_momentum	0,1	0	512	20	256-128	0 none	fp32	he		3	1,9005	0,2305	19	0,2180	0,2421	0,9088	0,8556	3,1746569		FALSE			figures/base-s3.png	GPU-memory=N/A on CPU.	-0,0454	0,0242	0,0008	Không
		Mixed precision may not improve end-to-end speed on this small MLP; numerical trajectories may change.																								figures/amp-bf16.png	CPU GPU-memory=N/A; unavailable: FP16 training requires CUDA; measured BF16 on CPU. Prediction: Mixed precision may not improve end-to-end speed on this small MLP; numerical trajectories may change. Wall-clock mean includes long execution interruption; median=65.670s. See REPORT.					
amp-bf16	amp		ce	sgd_momentum	0,1	0	512	20	256-128	0 none	bf16	he		1	2,2691	0,2345	18	0,2161	0,2391	0,9071	0,8500	2232,5321		FALSE			figures/batch-2048.png	GPU-memory=N/A on CPU. Prediction: Larger batch reduces updates per epoch and may slow convergence at fixed LR.	0,3232	0,0230	-0,0048	Không
batch-2048	hparam	Larger batch reduces updates per epoch and may slow convergence at fixed LR.	ce	sgd_momentum	0,1	0	2048	20	256-128	0 none	fp32	he		1	2,2691	0,2869	20	0,2745	0,2869	0,8822	0,8095	2,1558396		FALSE			figures/clip-high-none.png	GPU-memory=N/A on CPU. Prediction: Larger batch reduces updates per epoch and may slow convergence at fixed LR.	0,3232	0,0124	-0,0453	Có
		Ten-fold LR may cause large gradients, unstable loss, or divergence.																								figures/clip-high-none.png	GPU-memory=N/A on CPU. Prediction: Ten-fold LR may cause large gradients, unstable loss, or divergence. Clipped 0/14540 steps.					
clip-high-none	clipping		ce	sgd_momentum	1	0	512	20	256-128	0 none	fp32	he		1	2,2691	0,3628	20	0,3506	0,3628	0,8598	0,7441	3,5889091		FALSE			figures/clip-high-on.png	GPU-memory=N/A on CPU. Prediction: At the same high LR clipping may limit spikes, but cannot guarantee stable convergence.	0,3232	0,0122	-0,1107	Có
		At the same high LR clipping may limit spikes, but cannot guarantee stable convergence.																								figures/clip-high-on.png	GPU-memory=N/A on CPU. Prediction: At the same high LR clipping may limit spikes, but cannot guarantee stable convergence. Clipped 1986/14540 steps.					
clip-high-on	clipping		ce	sgd_momentum	1	0	512	20	256-128	0	0,287	fp32	he	1	2,2691	0,3014	19	0,2993	0,3180	0,8848	0,8091	3,214487		FALSE			figures/clip-normal.png	GPU-memory=N/A on CPU. Prediction: Clipping should activate at normal LR and may slow learning. Clipped 14538/14540 steps.	0,3232	0,0187	-0,0457	Có
		Clipping should activate at normal LR and may slow learning.																								figures/clip-normal.png	GPU-memory=N/A on CPU. Prediction: Clipping should activate at normal LR and may slow learning. Clipped 14538/14540 steps.					
clip-normal	clipping		ce	sgd_momentum	0,1	0	512	20	256-128	0	0,287	fp32	he	1	2,2691	0,2538	18	0,2376	0,2570	0,8983	0,8278	3,3540977		FALSE			figures/drop-0.1.png	GPU-memory=N/A on CPU. Prediction: Dropout may reduce the train-val gap but hurt a baseline still improving at epoch 20.	0,3232	0,0194	-0,0271	Có
		Dropout may reduce the train-val gap but hurt a baseline still improving at epoch 20.																								figures/drop-0.1.png	GPU-memory=N/A on CPU. Prediction: Dropout may reduce the train-val gap but hurt a baseline still improving at epoch 20.					
drop-0.1	dropout		ce	sgd_momentum	0,1	0	512	20	256-128	0,1 none	fp32	he		1	2,2691	0,2507	20	0,2420	0,2507	0,8982	0,8418	5,7497311		FALSE			figures/final-s1.png	GPU-memory=N/A on CPU. Prediction: Selected from opt-adam-0.003 by validation; extended 40 epochs.	0,3232	0,0087	-0,0131	Có
		Validation-selected configuration; extended 40-epoch budget																								figures/final-s1.png	GPU-memory=N/A on CPU. Selected from opt-adam-0.003 by validation; extended 40 epochs.					
final-s1	final		ce	adam	0,003	0	512	40	256-128	0 none	fp32	he		1	2,2691	0,1838	36	0,1665	0,2019	0,9284	0,8887	12,19513		FALSE			figures/final-s2.png	GPU-memory=N/A on CPU. Prediction: Selected from opt-adam-0.003 by validation; extended 40 epochs.	0,3232	0,0355	0,0339	Có
		Validation-selected configuration; extended 40-epoch budget																								figures/final-s2.png	GPU-memory=N/A on CPU. Prediction: Selected from opt-adam-0.003 by validation; extended 40 epochs.					
final-s2	final		ce	adam	0,003	0	512	40	256-128	0 none	fp32	he		2	1,9782	0,1787	40	0,1429	0,1787	0,9306	0,8950	14,718428		FALSE			figures/final-s3.png	GPU-memory=N/A on CPU. Prediction: Selected from opt-adam-0.003 by validation; extended 40 epochs.	0,0322	0,0359	0,0402	Có
		Validation-selected configuration; extended 40-epoch budget																								figures/final-s3.png	GPU-memory=N/A on CPU. Prediction: Selected from opt-adam-0.003 by validation; extended 40 epochs.					
final-s3	final		ce	adam	0,003	0	512	40	256-128	0 none	fp32	he		3	1,9005	0,1858	39	0,1503	0,1887	0,9294	0,8985	12,125354		FALSE	0,9282	0,8969	figures/init-normal.png	GPU-memory=N/A on CPU. Prediction: Selected from opt-adam-0.003 by validation; extended 40 epochs.	-0,0454	0,0384	0,0437	Có
		Small normal weights shrink activations and may delay convergence.																								figures/init-normal.png	GPU-memory=N/A on CPU. Prediction: Small normal weights shrink activations and may delay convergence.					
init-normal	init		ce	sgd_momentum	0,1	0	512	20	256-128	0 none	fp32	normal		1	1,9460	0,2470	20	0,2306	0,2470	0,9019	0,8493	5,8369033		FALSE			figures/init-xavier.png	GPU-memory=N/A on CPU. Prediction: Xavier may shrink ReLU activations and may delay convergence.	0,0001	0,0164	-0,0056	Có
		Xavier may shrink ReLU activations relative to He; a shallow MLP may still train well.																								figures/init-xavier.png	GPU-memory=N/A on CPU. Prediction: Xavier may shrink ReLU activations relative to He; a shallow MLP may still train					
init-xavier	init		ce	sgd_momentum	0,1	0	512	20	256-128	0 none	fp32	xavier		1	2,0222	0,2395	18	0,2225	0,2451	0,9037	0,8470	5,8190653		FALSE			figures/init-zeros.png	GPU-memory=N/A on CPU. Prediction: Zero initialization leaves hidden ReLU gradients zero; only output bias can learn class priors.	0,0763	0,0226	-0,0079	Có
		Zero initialization leaves hidden ReLU gradients zero; only output bias can learn class priors.																								figures/init-zeros.png	GPU-memory=N/A on CPU. Prediction: Zero initialization leaves hidden ReLU gradients zero; only output bias can learn class priors.					
init-zeros	init		ce	sgd_momentum	0,1	0	512	20	256-128	0 none	fp32	zeros		1	1,9459	1,2052	14	1,2066	1,2056	0,4876	0,0936	5,6398035		FALSE			figures/loss-mse.png	GPU-memory=N/A on CPU. Prediction: Zero initialization leaves hidden ReLU gradients zero; only output bias can learn class priors.	0,0000	-0,0010	-0,7612	Có
		MSE on raw logits versus one-hot targets may fit classification less effectively than CE.																								figures/loss-mse.png	GPU-memory=N/A on CPU. Prediction: MSE on raw logits versus one-hot targets may fit classification less effectively than CE. Raw-logit MSE vs one-hot; ln(7) step0-gap formula is only meaningful for CE.					
loss-mse	loss		mse	sgd_momentum	0,1	0	512	20	256-128	0 none	fp32	he		1	0,6359	0,0294	19	0,0290	0,0296	0,8711	0,7361	3,1543368		FALSE			figures/ir-0.01-s1.png	GPU-memory=N/A on CPU. Prediction: MSE on raw logits versus one-hot targets may fit classification less effectively than CE. Raw-logit MSE vs one-hot; ln(7) step0-gap formula is only meaningful for CE.	-1,3101	0,0007	-0,1188	Có
		Validation-only LR screening; 3 epoch budget																								figures/ir-0.01-s1.png	GPU-memory=N/A on CPU.					
lr-0.01-s1	lr_screen		ce	sgd_momentum	0,01	0	512	3	256-128	0 none	fp32	he		1	2,2691	0,4952	3	0,4962	0,4952	0,7915	0,5986	3,6781873		FALSE			figures/ir-0.03-s1.png	GPU-memory=N/A on CPU.	0,3232	-0,0010	-0,2563	Có
		Validation-only LR screening; 3 epoch budget																								figures/ir-0.03-s1.png	GPU-memory=N/A on CPU.					
lr-0.03-s1	lr_screen		ce	sgd_momentum	0,03	0	512	3	256-128	0 none	fp32	he		1	2,2691	0,4326	3	0,4322	0,4326	0,8191	0,6488	3,1437728		FALSE			figures/ir-0.1-s1.png	GPU-memory=N/A on CPU.	0,3232	0,0004	-0,2060	Có
		Validation-only LR screening; 3 epoch budget																								figures/ir-0.1-s1.png	GPU-memory=N/A on CPU.					
lr-0.1-s1	lr_screen		ce	sgd_momentum	0,1	0	512	3	256-128	0 none	fp32	he		1	2,2691	0,3776	3	0,3722	0,3776	0,8460	0,7062	3,3497178		FALSE			figures/opt-adam-0.001.png	GPU-memory=N/A on CPU. Prediction: Adam may converge faster through per-parameter adaptive updates.	0,3232	0,0054	-0,1486	Có
		Adam may converge faster through per-parameter adaptive updates.																								figures/opt-adam-0.001.png	GPU-memory=N/A on CPU. Prediction: Adam may converge faster through per-parameter adaptive updates.					
opt-adam-0.001	optimizer		ce	adam	0,001	0	512	20	256-128	0 none	fp32	he		1	2,2691	0,2454	18	0,2292	0,2495	0,9017	0,8459	4,0324602		FALSE			figures/opt-adam-0.003.png	GPU-memory=N/A on CPU. Prediction: Larger Adam LR may accelerate early fitting but increase oscillation.	0,3232	0,0203	-0,0090	Có
		Larger Adam LR may accelerate early fitting but increase oscillation.																								figures/opt-adam-0.003.png	GPU-memory=N/A on CPU. Prediction: Larger Adam LR may accelerate early fitting but increase oscillation.					
opt-adam-0.003	optimizer		ce	adam	0,003	0	512	20	256-128	0 none	fp32	he		1	2,2691	0,2106	20	0,1845	0,2106	0,9160	0,8740	4,1069464		FALSE			figures/opt-sgdm-0.03.png	GPU-memory=N/A on CPU. Prediction: SGD momentum LR 0.03 may converge more slowly than 0.1.	0,3232	0,0261	0,0192	Có
		SGD momentum LR 0.03 may converge more slowly than 0.1.																								figures/opt-sgdm-0.03.png	GPU-memory=N/A on CPU. Prediction: SGD momentum LR 0.03 may converge more slowly than 0.1.					
opt-sgdm-0.03	optimizer		ce	sgd_momentum	0,03	0	512	20	256-128	0 none	fp32	he		1	2,2691	0,2594	20	0,2450	0,2594	0,8961	0,8264	3,5128371		FALSE								